Theory of Computation	1-7	Rovlaw of Mathornatical Theory
Let us discuss each one by one-		
I) Negation		
horizontal bar on letter.		
	S	1s
	T	F
	F	T
S	1s	
T	F	
F	T	

i Conjunction AND

If A and B are two propositions then conjunction of A and B is denoted by A^B. The truth table for this connective is

A	B	AB
T	T	T
T	F	F
F	T	F
F	F	F

I) Disjunction (OR)

If A and B are two propositions then disjunction of A and B is denoted by AvB.The truth table for this connective is

A	B	AvB
T	T.	T
T	F	T
F	T	T
F	F	F

iv) Implication

If A and B are two propositions then if A then B is a proposition.It can be denoted byA-BorA=B.

The truth table for this connective is • Review of Mathematical Theorye

A		
	B	
		AB
T		
	T	
		T
T		
	F	F
F		
	T	T
F		
	F	T

For example: Find truth value for following proposition. If an apple is not green then it must be red A is false and B is true. Hence the proposition is true. vIf and only if f A and B are two propositions then A if and only if B. t can be denoted as A The truth table will be

A	B	AB
T	T	T
T	F	F
F	T	F
F	F	T

For example: I will go to the school if and only if it is not rainingHere statement A is I will go to the school. The statement B is It is not raining Thus the resultant proposition is false.

1.5 Relations

Relaonship is a major aspect between two objects even this is true in our real life. One ofect can be related with the other object by a some relation.Then those two objects forma pair based on this certain relationship.

Definition The relation R is a collction for the set S which represents the pair of elements.

For example:(a b) is in R.We can represent their relation as a R b. The first pair is chosen from a set called range.

Theory of Computation

Properties of Relations

A relation R on set S is

1. Reflexive if iRi for all i in S.
2. Irreflexive if iRi is false for all i in sS.

Transitive if iRj and jRk imply iRk. 4Symmetric if iRj implies jRi. 5Asymmetric if iRj implies that jRi is false Every asymmetric relation must be irreflexive.

For example: if A=ab then

refexive relation R can be=a, ab, b
irreflexive relation R can be=ab
transitive relation R can be=|a, b), b, aaa|
symmetric relation R can be=ab)b, a
asymmetric relation R can be =a b

Equivalent Relation

A relation is said to be equivalence relation if it is reflexive, symmetric and transitive, over some set S. Suppose R is a set of relations and S is the set of elements.

For example: S is the set of lines in a plane and R is the relation of lines intersecting to each other. A relation is said to be antisymmetric if a, b)e R and b ae R then a =a

Example 1.5: Determine whether R is equivalence relation or not where A=|0121.R=001, 0), 11)2, 2)21 SolutionThe R is reflexive because 0, 0)1, 1)2.2E R. Where as R is not symmetric because (0, 1) is not in R whereas (1, 0) is in R. Hence the R is not a equivalence relation.

Closures of Relatlons

Sometimes when the relation R is given, it may not be reflexive or transitive By adding some pairs we make the relation as reflexive or transitive. For exampleLet(ab)b, c) a, a)bb